ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS5511 ARTIFICIAL INTELLIGENCE

FINAL EXAMINATION

DATE: 29TH MAY 2025

TIME: 14:00HRS – 17:00HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

(Total: 50 marks)

Artificial Intelligence is revolutionizing the way in which different companies across compete and grow across the world by representing a new production factor that can drive business profitability

•Big data science and increased processing capacity has increased use cases for interoperability of different systems making advanced systems simpler to use

In very simple terms, AI has turned R&D into a needed business asset that helps in one or all of the following ways.

- Improved Efficiency
- Cost savings
- Reduced Human error potential –full automation or decision support
- Predict and model future scenarios therefore limiting risk for unexpected problems

Intelligent agents are a fundamental building block of artificial intelligence.

a) Define and briefly describe the 5 (five) main categories of Intelligent agents (30 marks)
b) State and describe 4 real world implementations of rational agents (20 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

(25 marks)

Whenever an agent is confronted with a problem, its first action is to seek a solution in its knowledge system. This is known as the search for the solution in the knowledge base. Another way to search for a solution is by going into a different state.

In artificial intelligence, problems can be solved by using searching algorithms, evolutionary computations, knowledge representations, etc.

a) Define and briefly describe the steps that are involved in using search for problem solving? (10 marks)
b) What are the core properties of search algorithms? (8 marks)
c) Distinguish between Uninformed and Informed search algorithms? (1 mark)
d) Give 3 examples of Uninformed search algorithms and state at least one advantage and one disadvantage of each? (6 marks)

Question 3

(25 marks)

A rational agent is a mathematical model that tries to represent the behaviour of an intelligent being, like a person or animal. It uses a set of rules to determine the best course of action for a given situation.

These agents usually work by comparing their current state with their previous state and then choosing an action based on how much better or worse they feel about their position now compared to before.

For example, if you're hungry, you might want to eat something. If you're not hungry any longer, you might stop eating. A rational agent will do this repeatedly until it reaches some goal or decides it's time for bed (or both).

a) State and describe any 5 areas in which rational agents differ from Intelligent agents (10 marks)
b) State and briefly describe 3 ways in which knowledge can be represented (15 marks)

Question 4

(25 marks)

In machine learning, what is the distinguishing characteristic between supervised and unsupervised learning?

a) State and describe at least two techniques of the heuristic search method? (15 marks)
b) What is adversarial search methodology and where is it mostly applied? (10 marks)

END OF EXAMINATION